

32-bit microcontroller

FLASH erase and write protection function

Application Note

Rev1.00 January 2025

Applicable objects

Product series	Product model	Product series	Product model
L Series	HC32F003	F Series	HC32L110
	HC32F005		HC32F030
	HC32L130		HC32F072
	HC32L136		HC32F170
	HC32L072		HC32F176
	HC32L073		HC32F190
	HC32L170		HC32F196
	HC32L176		HC32F002
	HC32L190		HC32F052
	HC32L196		HC32F420
	HC32L166		
	HC32L186		

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1 Overview

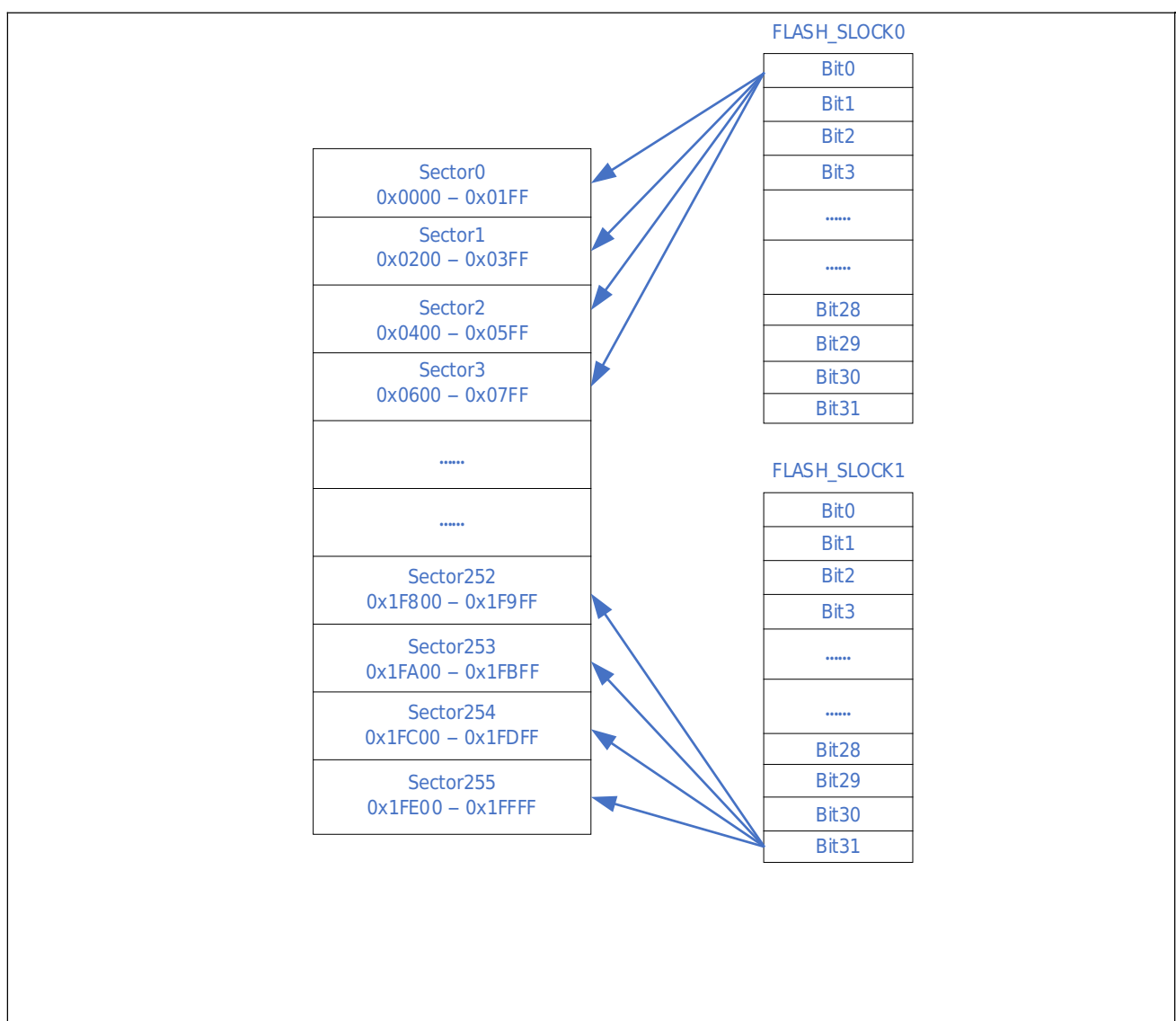
This document mainly introduces some precautions for using FLASH erase and write in the program of Xiaohua Semiconductor's specific series MCU.

2 FLASH erase protection

2.1 FLASH erase protection register

The FLASH module of the MCU series described in this article has an erase protection register. When the control bit of the register is 0, the corresponding page of FLASH can be erased and written. The size of each FLASH page is 512 bytes. The whole chip is divided into several pages. Every 4 pages share an erase protection control bit of the erase protection register. Each series of MCU has 1~4 erase protection registers SLOCKx according to the size of FLASH.

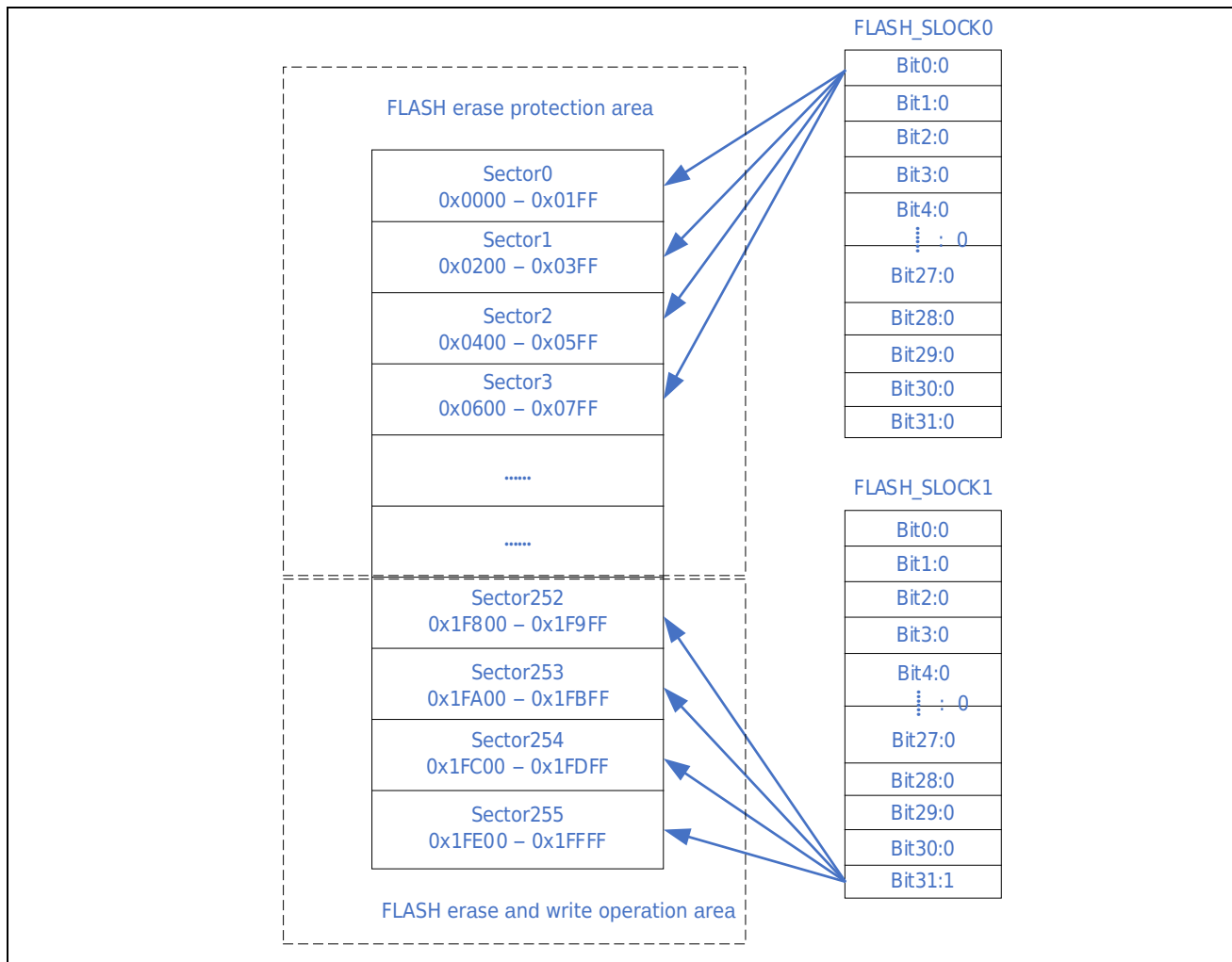
For example, the FLASH size in the figure below is 128KB, the sectors it has are 0~255, and the erase protection registers it has are FLASH_SLOCK0 and FLASH_SLOCK1.



2.2 Use of erase protection register

When the program needs to erase or write certain pages of FLASH, the control bit of the erase protection register corresponding to the page needs to be set to 1 first: erase and write are allowed.

When releasing the erase protection and allowing erase and write (unlocking), it is strongly recommended that users only release the control bit corresponding to the page that needs to be erased and written, and the control bit of the erase protection register SLOCKx corresponding to the page in the user code area should never be set to 1.



When rewriting the erase protection register, it is recommended to check the correctness of the parameters written to the register. If the passed parameters are incorrect, the operation will not be performed and an error flag will be returned.

It is also recommended to check the correctness of the operation address during erase and write operations. If the address is not in the correct range, the operation will not be performed and an error flag will be returned.

The operation of the SLOCKx register also requires the BYPASS register to be operated first. Therefore, when the user operates these registers, if the interrupt is not masked, the value of the corresponding register needs to be queried after the operation is completed to see if the write is successful. If the write is unsuccessful, it is necessary to try to repeat the operation. It is also recommended to add a limit on the number of repeated operations to avoid the program being stuck due to continuous failure of the operation.

In addition, the following register operations also require the BYPASS register to be operated first, so after the operation is completed, it is also necessary to query the readback confirmation and retry operations as mentioned above:

Parameter register, CR, WAIT (some series exist) and CACHE (some series exist).

3 Summary

The above chapters briefly introduce the application methods of the FLASH erase and write protection function of Xiaohua Semiconductor MCU*. In the actual application development process, if the user needs to further understand the use and operation of this module, the corresponding user manual should prevail. The samples and driver libraries mentioned in this chapter can be used as further experiments and learning for users, and can also be used as references in actual development.

Version Revision History

Version	Revision Date	Content
Rev1.00	2025-1-20	Initial version released.